

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Data Warehousing and Data Mining

COURSE CODE: CSA1609

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)

Assessment Tool Weightage: 30%

QUESTIONS: 3

Total Marks:30

Annexure C

Case study

Criteria	Understanding of Case (2.5)	Application of Concepts (2.5)	Analysis (2)	Solution Design (2)	Clarity (1)	Total(10)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						

Code of Conduct:

IG Madhulatha(192311295) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

G Madhulatha

ANNEXURE A

Case Study:

1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them.

One night, the shepherd boy sees a real wolf approaching the flock and calls out, "Wolf!" The villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock into lamb chops. The town goes hungry. Panic ensues. Identify TP,FP,FN,TN

2. Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images. Identify TP,FP,FN,TN and Calculate the Precision and Recall and F1 Score .

3. An email service provider wants to classify emails as **Spam or Not Spam** using features:

- Contains "Offer" (Yes/No)
- Contains "Free" (Yes/No)
- Email Length (Short/Long)

Sample Dataset

Offer	Free	Length	Class
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Questions

1. Calculate prior probabilities:
 - $P(\text{Spam})$, $P(\text{Not Spam})$
2. Compute conditional probabilities for each feature.
3. Using **Naïve Bayes**, classify:
 - (Offer=Yes, Free=No, Length=Short)
4. Compare results using **WEKA (Naïve Bayes classifier)**.
5. Discuss the **independence assumption** in Naïve Bayes

ANSWERS

1. Shepherd Boy and Wolf – Identify TP, FP, FN, TN

First, consider:

- **Positive = Wolf is present**
- **Negative = Wolf is not present**
- **Prediction = Boy says “Wolf!” / villagers believe there is a wolf**

A confusion matrix has four possible outcomes:

Term	Meaning
TP – True Positive	Wolf is actually present and an alarm is correctly raised/believed
FP – False Positive	Wolf is not present, but an alarm says wolf is present
FN – False Negative	Wolf is actually present, but the alarm is ignored/not believed
TN – True Negative	Wolf is not present and no wolf alarm is raised

First Situation

The shepherd boy cries:

“Wolf!”

But actually, **there is no wolf**.

- Predicted: Wolf present → **Positive**
- Actual: No wolf → **Negative**

Therefore:

False Positive (FP)

The boy gave an alarm even though there was no wolf.

So,

FP = 1

Second Situation

Later, a **real wolf appears**.

The boy cries:

“Wolf!”

But the villagers do not believe him and assume there is no wolf.

- Actual: Wolf present → **Positive**
- Villagers' decision: No wolf / ignore alarm → **Negative**

Therefore:

False Negative (FN)

A real danger existed, but it was treated as if there was no danger.

So,

FN = 1

What about TP and TN?

There is no situation in the story where:

- a wolf was actually present and correctly acted upon → **TP**
- no wolf was present and correctly identified as no wolf → **TN**

Therefore:

Situation	Actual	Prediction/Decision	Result
Boy falsely cries “Wolf!”	No Wolf	Wolf	FP
Real wolf appears but villagers don't believe	Wolf	No Wolf	FN

Final Answer

$TP = 0$
$FP = 1$
$FN = 1$
$TN = 0$

Interpretation

The story explains why repeated **false positives** can be dangerous. Because the shepherd boy repeatedly gives false alarms, the villagers lose trust. Eventually, this leads to a **false negative** when a real wolf appears and the villagers fail to respond.

2. Cat Image Classification

Given Information

Total images:

100

Images containing cats:

30

Images without cats:

$$100 - 30 = 70$$

The model correctly predicts cats in **25 out of 30 cat images**.

Therefore:

True Positive

Cat is actually present and model predicts cat.

$$\boxed{TP = 25}$$

Since there are 30 cat images:

$$FN = 30 - 25$$

$$\boxed{FN = 5}$$

Now, among the **70 images without cats**, the model correctly predicts absence of a cat in **50 images**.

Therefore:

$$\boxed{TN = 50}$$

Remaining non-cat images are incorrectly predicted as cat.

$$FP = 70 - 50$$

$$\boxed{FP = 20}$$

Confusion Matrix

	Predicted Cat	Predicted No Cat
Actual Cat	TP = 25	FN = 5
Actual No Cat	FP = 20	TN = 50

Therefore:

$$\boxed{TP = 25, FP = 20, FN = 5, TN = 50}$$

Check:

$$25 + 20 + 5 + 50 = 100$$

Correct.

Precision

Precision tells us:

Out of all images predicted as cats, how many actually contain cats?

Formula:

$$Precision = \frac{TP}{TP + FP}$$

Substitute the values:

$$\begin{aligned} Precision &= \frac{25}{25 + 20} \\ &= \frac{25}{45} \\ &= 0.5556 \end{aligned}$$

Therefore:

$$\boxed{Precision = 55.56\%}$$

Interpretation

When the model predicts that an image contains a cat, it is correct approximately **55.56% of the time**.

Recall

Recall tells us:

Out of all actual cat images, how many cats were correctly detected?

Formula:

$$Recall = \frac{TP}{TP + FN}$$

Substitute:

$$\begin{aligned}
 Recall &= \frac{25}{25 + 5} \\
 &= \frac{25}{30} \\
 &= 0.8333
 \end{aligned}$$

Therefore:

$$\boxed{Recall = 83.33\%}$$

Interpretation

The model detects approximately **83.33% of all actual cat images**.

F1 Score

F1 Score is the harmonic mean of **Precision and Recall**.

Formula:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

We have:

$$\begin{aligned}
 Precision &= 0.5556 \\
 Recall &= 0.8333
 \end{aligned}$$

Therefore:

$$F1 = 2 \times \frac{0.5556 \times 0.8333}{0.5556 + 0.8333}$$

First multiply:

$$0.5556 \times 0.8333 \approx 0.463$$

Then:

$$0.5556 + 0.8333 = 1.3889$$

Therefore:

$$\begin{aligned}
 F1 &= 2 \times \frac{0.463}{1.3889} \\
 F1 &\approx 0.6667
 \end{aligned}$$

Therefore:

$$F1\ Score = 66.67\%$$

Final Answer for Question 2

Measure	Value
TP	25
FP	20
FN	5
TN	50
Precision	55.56%
Recall	83.33%
F1 Score	66.67%

3. Email Spam Classification Using Naïve Bayes

Given Dataset

Email Offer Free Length Class

1	Yes	Yes	Short	Spam
2	Yes	No	Long	Not Spam
3	No	Yes	Short	Spam
4	No	No	Long	Not Spam
5	Yes	Yes	Long	Spam
6	No	No	Short	Not Spam

Total emails:

6

Spam emails:

3

Not Spam emails:

3

3.1 Calculate Prior Probabilities

Prior probability represents the probability of a class before considering any features.

P(Spam)

There are 3 Spam emails out of 6.

$$P(\text{Spam}) = \frac{3}{6}$$

$P(\text{Spam}) = 0.5$

or

50%

P(Not Spam)

There are 3 Not Spam emails out of 6.

$$P(\text{NotSpam}) = \frac{3}{6}$$

$P(\text{NotSpam}) = 0.5$

or

50%

Therefore:

Class	Prior Probability
Spam	0.5
Not Spam	0.5

3.2 Conditional Probabilities

Now calculate probabilities of each feature value given the class.

A. For Spam Emails

Spam records are:

Offer Free Length

Yes Yes Short

No Yes Short

Yes Yes Long

Total Spam:

Offer Feature

Offer = Yes appears in 2 Spam emails.

$$P(\text{Offer} = \text{Yes} \mid \text{Spam}) = \frac{2}{3}$$

0.667

Offer = No appears once.

$$P(\text{Offer} = \text{No} \mid \text{Spam}) = \frac{1}{3}$$

0.333

Free Feature

All three Spam emails contain Free = Yes.

$$P(\text{Free} = \text{Yes} \mid \text{Spam}) = \frac{3}{3} = 1$$

Therefore:

$P(\text{Free} = \text{Yes} \mid \text{Spam}) = 1$

There are no Spam emails with Free = No.

$$P(\text{Free} = \text{No} \mid \text{Spam}) = \frac{0}{3}$$

$P(\text{Free} = \text{No} \mid \text{Spam}) = 0$

Length Feature

Two Spam emails are Short.

$$P(\text{Length} = \text{Short} \mid \text{Spam}) = \frac{2}{3}$$

0.667

One Spam email is Long.

$$P(\text{Length} = \text{Long} \mid \text{Spam}) = \frac{1}{3}$$

0.333

Spam Conditional Probability Table

Feature	Probability
P(Offer=Yes Spam)	$2/3 = 0.667$
P(Offer=No Spam)	$1/3 = 0.333$
P(Free=Yes Spam)	$3/3 = 1$
P(Free=No Spam)	$0/3 = 0$
P(Length=Short Spam)	$2/3 = 0.667$
P(Length=Long Spam)	$1/3 = 0.333$

B. For Not Spam Emails

Not Spam records are:

Offer Free Length

Yes	No	Long
No	No	Long
No	No	Short

Total:

3

Offer Feature

Offer = Yes occurs once.

$$P(\text{Offer} = \text{Yes} \mid \text{NotSpam}) = \frac{1}{3}$$

0.333

Offer = No occurs twice.

$$P(\text{Offer} = \text{No} \mid \text{NotSpam}) = \frac{2}{3}$$

0.667

Free Feature

Free = Yes occurs zero times.

$$P(\text{Free} = \text{Yes} \mid \text{NotSpam}) = 0$$

Free = No occurs three times.

$$P(\text{Free} = \text{No} \mid \text{NotSpam}) = \frac{3}{3}$$

1

Length Feature

Length = Short occurs once.

$$P(\text{Length} = \text{Short} \mid \text{NotSpam}) = \frac{1}{3}$$

0.333

Length = Long occurs twice.

$$P(\text{Length} = \text{Long} \mid \text{NotSpam}) = \frac{2}{3}$$

0.667

Not Spam Conditional Probability Table

Feature	Probability
P(Offer=Yes Not Spam)	1/3 = 0.333
P(Offer=No Not Spam)	2/3 = 0.667
P(Free=Yes Not Spam)	0
P(Free=No Not Spam)	1
P(Length=Short Not Spam)	1/3 = 0.333
P(Length=Long Not Spam)	2/3 = 0.667

3.3 Classify the New Email Using Naïve Bayes

We have to classify:

Offer = Yes
Free = No
Length = Short

We need to compare:

$$P(\text{Spam} \mid \text{YesNoShort})$$

and

$$P(NotSpam \mid YesNoShort)$$

Naïve Bayes is based on Bayes' theorem:

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For classification, we compare the unnormalized scores:

$$P(Class) \times P(Offer \mid Class) \times P(Free \mid Class) \times P(Length \mid Class)$$

Probability for Spam

$$\begin{aligned} P(Spam) &= 0.5 \\ P(Offer = Yes \mid Spam) &= \frac{2}{3} \\ P(Free = No \mid Spam) &= 0 \\ P(Length = Short \mid Spam) &= \frac{2}{3} \end{aligned}$$

Therefore:

$$Score(Spam) = 0.5 \times \frac{2}{3} \times 0 \times \frac{2}{3}$$

Because one probability is zero:

$$\boxed{Score(Spam) = 0}$$

Probability for Not Spam

We have:

$$\begin{aligned} P(NotSpam) &= 0.5 \\ P(Offer = Yes \mid NotSpam) &= \frac{1}{3} \\ P(Free = No \mid NotSpam) &= 1 \\ P(Length = Short \mid NotSpam) &= \frac{1}{3} \end{aligned}$$

Therefore:

$$\begin{aligned} Score(NotSpam) &= 0.5 \times \frac{1}{3} \times 1 \times \frac{1}{3} \\ &= \frac{1}{2} \times \frac{1}{9} \end{aligned}$$

$$= \frac{1}{18}$$

$$\boxed{Score(NotSpam) = 0.0556}$$

Now compare:

$$Spam = 0$$

$$NotSpam = 0.0556$$

Since:

$$0.0556 > 0$$

Final Classification

NOT SPAM

Therefore, the email

(Offer = Yes, Free = No, Length = Short)

will be classified as:

NOT SPAM

Zero Probability Problem

Notice that:

$$P(Free = No \mid Spam) = 0$$

Because of this single zero value, the entire Spam probability becomes zero.

This is called the **zero-frequency problem** in Naïve Bayes.

A common solution is **Laplace smoothing**.

Classification Using Laplace Smoothing

For a binary feature:

$$P(x \mid Class) = \frac{Count + 1}{Total + 2}$$

because every feature has two possible values.

For Spam:

Offer = Yes

$$\begin{aligned}P(\text{Offer} = \text{Yes} \mid \text{Spam}) &= \frac{2 + 1}{3 + 2} \\&= \frac{3}{5} = 0.6\end{aligned}$$

Free = No

$$\begin{aligned}P(\text{Free} = \text{No} \mid \text{Spam}) &= \frac{0 + 1}{3 + 2} \\&= \frac{1}{5} = 0.2\end{aligned}$$

Length = Short

$$\begin{aligned}P(\text{Short} \mid \text{Spam}) &= \frac{2 + 1}{3 + 2} \\&= \frac{3}{5} = 0.6\end{aligned}$$

Thus:

$$\begin{aligned}\text{Score}(\text{Spam}) &= 0.5 \times 0.6 \times 0.2 \times 0.6 \\ \boxed{\text{Score}(\text{Spam})} &= 0.036\end{aligned}$$

Not Spam with Laplace Smoothing

Offer = Yes

$$\begin{aligned}P(\text{Offer} = \text{Yes} \mid \text{NotSpam}) &= \frac{1 + 1}{3 + 2} \\&= \frac{2}{5} = 0.4\end{aligned}$$

Free = No

$$\begin{aligned}P(\text{Free} = \text{No} \mid \text{NotSpam}) &= \frac{3 + 1}{3 + 2} \\&= \frac{4}{5} = 0.8\end{aligned}$$

Length = Short

$$P(\text{Short} \mid \text{NotSpam}) = \frac{1 + 1}{3 + 2}$$

$$= \frac{2}{5} = 0.4$$

Therefore:

$$Score(NotSpam) = 0.5 \times 0.4 \times 0.8 \times 0.4$$

$$\boxed{Score(NotSpam) = 0.064}$$

Compare:

Class	Score
Spam	0.036
Not Spam	0.064

Since:

$$0.064 > 0.036$$

the result is again:

NOT SPAM

So both the basic calculation and Laplace-smoothed calculation classify the new email as **Not Spam**.

3.4 Comparison Using WEKA Naïve Bayes

The dataset can be entered into WEKA using an **ARFF file**.

An example dataset is:

@relation email_spam

@attribute Offer {Yes,No}

@attribute Free {Yes,No}

@attribute Length {Short,Long}

@attribute Class {Spam,NotSpam}

@data

Yes,Yes,Short,Spam

Yes,No,Long,NotSpam

No,Yes,Short,Spam

No,No,Long,NotSpam

Yes,Yes,Long,Spam

No,No,Short,NotSpam

For the unknown instance:

Yes,No,Short,?

Steps in WEKA

1. Open **WEKA**.
2. Click **Explorer**.
3. Select **Preprocess**.
4. Click **Open file**.
5. Load the .arff file.
6. Select the **Class** attribute as the target variable.
7. Go to the **Classify** tab.
8. Click **Choose**.
9. Select:

bayes → NaiveBayes

10. Select an appropriate testing option, such as **Use training set** or load a supplied test set.
11. Click **Start**.
12. WEKA will display the classification results.

For the new instance:

Offer = Yes, Free = No, Length = Short

we would expect it to be classified as:

Not Spam

The exact probability values displayed by WEKA can differ from the simple hand calculation because WEKA's implementation uses probability estimators and smoothing techniques rather than simply allowing every unseen feature/class combination to remain at zero.

Manual vs WEKA Comparison

Aspect	Manual Naïve Bayes	WEKA
Prior probabilities	Calculated manually	Calculated automatically
Conditional probabilities	Calculated manually	Calculated automatically
Zero probabilities	Can occur	Handled through estimators/smoothing
Classification	Not Spam	Expected: Not Spam
Dataset size	6	6
Ease of use	More calculations	Automatic
Best use	Understanding algorithm	Practical ML experiments

3.5 Independence Assumption in Naïve Bayes

The most important assumption in Naïve Bayes is that the features are **conditionally independent given the class**.

For our example, the features are:

- Offer
- Free
- Length

Naïve Bayes assumes that once we know whether an email is Spam or Not Spam, knowing that it contains “**Offer**” does not affect the probability that it contains “**Free**”, and neither affects its length.

Mathematically:

$$P(OfferFreeLength \mid Spam)$$

is simplified as:

$$P(Offer \mid Spam) \times P(Free \mid Spam) \times P(Length \mid Spam)$$

This greatly simplifies the calculations.

Example of Independence

Suppose an email is Spam.

Naïve Bayes treats:

Contains “Offer”

and

Contains “Free”

as independent pieces of information.

It calculates them separately:

$$P(\text{Offer} = \text{Yes} \mid \text{Spam})$$

and

$$P(\text{Free} = \text{Yes} \mid \text{Spam})$$

Then multiplies them.

Is This Assumption Realistic?

Not always.

In real spam emails, words such as:

“Free Offer”

may frequently appear together.

Therefore, the features **Offer** and **Free** may actually be related.

For example:

“Get your FREE OFFER today!”

If an email contains the word **Offer**, it may be more likely to contain the word **Free**.

So the features are not completely independent.

Nevertheless, Naïve Bayes often performs well even when the independence assumption is not perfectly satisfied.

Advantages of Independence Assumption

The independence assumption makes Naïve Bayes:

- computationally simple,
- fast to train,
- fast to predict,
- effective for high-dimensional data,
- suitable for text classification,

- suitable for spam detection.

It does not need to calculate every possible combination of features.

Disadvantages

The major limitation is that real-world features may be correlated.

For example:

Offer ↔ *Free*

The words may frequently occur together.

Naïve Bayes ignores such relationships.

As a result, it may sometimes overestimate or underestimate probabilities.

Final Summary

Question 1

Measure Value

TP	0
FP	1
FN	1
TN	0

The false “Wolf!” alarm is a **False Positive**, while ignoring the real wolf represents a **False Negative**.

Question 2

Measure Result

TP	25
FP	20
FN	5
TN	50
Precision	55.56%
Recall	83.33%
F1 Score	66.67%

Question 3

Prior probabilities:

$$P(\text{Spam}) = 0.5$$

$$P(\text{NotSpam}) = 0.5$$

For:

$$(Offer = Yes \text{ Free} = No \text{ Length} = Short)$$

without smoothing:

$$P(\text{Spam}) \text{ score} = 0$$

$$P(\text{NotSpam}) \text{ score} = 0.0556$$

Therefore:

$$\text{Prediction} = \text{NOT SPAM}$$

With Laplace smoothing:

$$\text{Spam} = 0.036$$

$$\text{NotSpam} = 0.064$$

So the prediction remains:

$$\text{NOT SPAM}$$

The Naïve Bayes independence assumption says that the features are treated as **independent given the class**, even though in real email data features such as “**Offer**” and “**Free**” may be related.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand